Training-Free Color Recovery for High-Density Multiplexed QR Codes

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***Abstract*—[TODO: draft; finalize after the introduction] High-capacity color QR codes stack three binary layers in the RGB channels, and decoding them depends on recovering each module's color from a phone photograph. We present a color recovery method that needs no offline training. It inverts a one-parameter linear leak between neighboring modules and then classifies colors with regularized discriminant analysis, fitted per photograph on the function patterns whose colors the symbol fixes. Both parameters were chosen on each of two datasets separately, and the chosen values are best or statistically tied with the best on both: the public CUHK-CQRC dataset of HiQ codes and a new dataset of 3428 photographs, which we release. On CUHK-CQRC the method decodes 42.9% of localized symbols, against 35.4% without leak inversion and 19.9% for nearest-centroid classification. On HiQ's own test split our open-source decoder reads 50.1% of the symbols that both it and the released HiQ decoder localize, against 29.5% for HiQ.**

***Index Terms*—color QR code, high-capacity barcode, color recovery, discriminant analysis, mobile imaging**

I. Introduction

A QR code stores one bit per module [1]. Printing three binary layers in the color channels of one symbol triples the data in the same area: each module shows one of eight colors, and its red, green and blue bits belong to the three layers [2]–[4]. Polychrome codes are also standardized, as JAB Code [5]. A multiplexed QR symbol keeps the layout of a standard symbol, so once the module colors are known each layer is read with standard Reed-Solomon decoding.

Recovering those colors from a phone photograph is the difficult step. Illumination and white balance shift every sample, overlapping colorant and sensor spectra mix the channels within a module, and lens blur, ink spread and JPEG chroma subsampling mix each module with its neighbors [4]. The last effect grows with density. Blur stays roughly fixed in pixels while modules shrink, so in a dense symbol a larger share of each sample comes from neighboring modules. Small module error rates also matter: a symbol fails as soon as one Reed-Solomon block in one layer holds more errors than it can correct. In our experiments two classifiers that differ by three points in module accuracy differ by more than a factor of two in symbols decoded (Section VI).

Earlier decoders handle these effects in several ways. Blasinski et al. estimate cross-channel mixing per image from pilot blocks or by expectation maximization [3], [6]. HiQ trains color classifiers offline on labelled photographs and learns cross-module interference terms with them [4]. Bagherinia and Manduchi decode blurred patches by iterative joint estimation [7], and screen-to-camera systems train networks end to end [8], [9]. A trained decoder depends on the printers, displays and phones in its training set. We look for a method that needs no training data, runs in time linear in the number of modules, and holds up on dense symbols.

Our method fits each photograph on its own. The finder, alignment and timing patterns of a multiplexed QR symbol have fixed colors, so they label a few hundred samples per photograph. Before fitting, a closed-form inverse of a symmetric leak between edge-adjacent modules, with one fixed fraction $\beta = 0.06$, reduces cross-module interference. The corrected samples are then classified by regularized discriminant analysis with shrink $\lambda = 0.5$, fitted to the labelled samples. On the public CUHK-CQRC dataset the method decodes 42.9% of localized symbols, against 35.4% without leak inversion and 19.9% for nearest-centroid classification; on a new dataset of qrism symbols it decodes 33.9%. On HiQ's own test split, our decoder reads 50.1% of the symbols that both it and the released HiQ decoder localize, against 29.5% for HiQ.

The contributions are:

- a training-free color recovery method for multiplexed QR codes: per-photograph calibration from function patterns, a one-parameter inverse of inter-module leakage, and regularized discriminant analysis (Section IV);
- a dataset of 3428 photographs of qrism symbols at seven versions, four error-correction levels and four phones, with the source render of every symbol;
- an evaluation on two datasets in which both parameters are chosen on each dataset and checked on the other, with paired tests, and an end-to-end comparison with the released HiQ decoder on its test split that also corrects a defect in its CMI step (Sections V and VI);
- an open-source Rust implementation. [TODO: repository URL]

Section II reviews related work, Section III describes the symbol designs and decoding pipeline, Section IV the method, Sections V and VI the experiments, and Section VII the limitations.

II. Related Work

A. Color and multiplexed 2D codes

Color has been added to 2D codes in two ways. Palette codes give each module one of several colors and carry

reference colors in the symbol: Microsoft's HCCB [10], HCC2D, which adds palette modules to a QR-style symbol [11], HD2DC [12], and JAB Code, which stores its palette redundantly and is standardized as ISO/IEC 23634 [5], [13]. Recent JAB Code work calibrates the printed colors to the printer [14] and improves version detection for large symbols [15]. Multiplexed codes instead stack binary layers, one per colorant or color channel [3], [16], so each layer is a standard symbol. André and Ferreira multiplexed three and six QR codes in CMY and modeled the recognition rate on the color constellation [2]; ColorQR streams RGB-multiplexed QR codes from a phone screen and splits each received frame into its channels [17]; El-Sheref et al. fold up to 24 layers into RGB shades [18]; Huang et al. design multiplexed codes whose plain layer a standard reader can decode [19], [20]. Other recent schemes separate the layers with narrowband illumination [21] or assign payload, metadata and signature to the three channels [22]. Most of these works report capacity or small capture tests. Of the multiplexed QR designs we found, HiQ is the only one evaluated on a large public set of phone photographs, CUHK-CQRC [4], [23]; public photographs of other color barcodes also exist [24]. Sirmen and Üstündağ survey color barcodes and their sources of distortion [25].

B. Color recovery

Decoding a color code means assigning each sampled module a color despite illumination, white balance and channel mixing. Palette codes compare samples with the reference colors printed in the symbol [11], [13]; Bagherinia and Manduchi analyze when colors can be separated without a palette [26]; Ong et al. adapt thresholds and color spaces to the lighting [27]. For per-colorant codes, Blasinski et al. model cross-channel interference as a linear mixing of colorant densities and estimate it per image, either from pilot blocks built into the synchronization markers or by expectation maximization over the data [3], [6]. HiQ trains its classifiers offline: QDA and a layered SVM, each optionally with a cross-module interference (CMI) term, are fitted to labelled CUHK-CQRC photographs [4]. Benito-Altamirano et al. work in the opposite direction and embed color charts in back-compatible QR codes to correct the colors of the scene [28].

Our calibration is closest to the pilot-block estimate of Blasinski et al.: both fit the color model to each photograph from modules whose colors are known. We differ in the model and in where the labels come from. We fit class-conditional Gaussians with shrunk covariances [29], which absorb illumination and channel mixing without a colorant model, and we take the labels from the finder, alignment and timing patterns that a multiplexed QR symbol already contains, so no symbol area goes to pilots and no photographs are needed for training.

C. Blur and inter-module interference

When modules shrink toward the size of the optical and demosaicing blur, each sampled module mixes with its neighbors. Bagherinia and Manduchi model a blurred patch as a linear combination of its own color and its neighbors' colors and decode all patches jointly by iteratively minimizing the observation error [7]. HiQ named the effect cross-module interference in high-density color QR codes and learns neighbor weights jointly with its classifier [4]. For monochrome QR codes, blind deblurring estimates a blur kernel in the pixel domain: from the finder patterns as a known signal [30], from priors on the finder and alignment patterns of each quadrant [31], or by a multi-scale search that stops once the code reads [32]. Channel models for print-to-camera and screen-to-camera links estimate spatial and color distortion from training symbols [33] or equalize color crosstalk with an MMSE filter on a transform-domain code [34].

Our leak model (1) is a linear neighbor model of the same kind as Bagherinia and Manduchi's, with four edge neighbors and one shared weight. Fixing that weight gives a closed-form inverse on the module grid, computed in one pass with no kernel estimate and no iteration over the symbol. We did not find 2020 to 2026 work that applies a fixed one-parameter grid inverse of this kind to color codes.

D. Learned decoders and screen-to-camera codes

Deep networks are now common for barcode localization and are used for decoding as well [35]. Liao et al. train CNNs on pairs of adjacent modules to read laser-marked Data Matrix symbols [36]. Most learned color decoders come from screen-to-camera and optical camera communication, where the transmitter is a display: Deep D2C-Net and DeepLight decode data embedded in displayed images with trained networks [8], [9], DeepCCB-OCC applies deep learning to complementary color barcodes [37], and CNN demodulators and neural equalizers read color shift keying with 4 to 512 colors [38], [39]. Model-based display codes use per-channel encoding [40] or temporal color vibration [41]. These systems are trained or tuned for a display and camera and often combine several frames. Our setting is one photograph of a printed or displayed static symbol taken with an unknown phone, and our method uses no training data. We found no learned decoder for multiplexed color QR codes in this period that reports results on CUHK-CQRC.

III. Multiplexed QR Symbols

Both symbol designs studied here stack three binary QR layers, one per RGB channel, so that each module shows one of eight colors: the color's red, green and blue bits are the module's bits in the three layers. A layer bit of 1 is drawn with its channel off, which matches the dark-module convention of a monochrome QR code [1].

HiQ [4] encodes three independent messages, one standard QR code per layer, so each layer can be decoded on its own once the colors are recovered.

qrism, the design implemented in the open-source library this work builds on, encodes one message. Its codeword stream is split into three equal parts, each written to one channel with its own Reed-Solomon blocks, so a symbol decodes only when all three channels do. The finder patterns use the same

complementary color pairs as HiQ (green ring and magenta stone top left, red and cyan top right, blue and yellow bottom left). Alignment patterns, timing patterns, and format and version information are black and white, as in a monochrome QR code.

A. Decoding pipeline

A decoder for either design runs four stages. (1) *Localization* finds the finder and alignment patterns and fits a homography per region of the symbol. In the library used here this runs on a min-channel projection of the photograph, which keeps colored finder rings dark, with sub-pixel finder and alignment centers. (2) *Sampling* reads one pixel at the projected center of every module. (3) *Color recovery* assigns each sample one of the eight colors. (4) *Decoding* splits the color grid into its three layers and runs standard Reed-Solomon decoding on each. This paper is about stage (3); the experiments in Section V hold stages (1), (2) and (4) fixed so that only color recovery varies.

IV. Color Recovery

Let $\mathbf{m}_p \in \mathbb{R}^3$ be the RGB value sampled at module p of an $N \times N$ grid and $c_p \in \{0, \dots, 7\}$ its color. The sample departs from the printed or displayed color in three ways [4]: illumination and white balance shift and scale all samples of a photograph; cross-channel interference mixes the layers within a module, because colorant spectra and camera sensitivities overlap; and cross-module interference mixes each module with its neighbors through lens blur, ink spread and the chroma subsampling of JPEG compression. The method handles the first two by fitting a class-conditional Gaussian model to each photograph, and the third by undoing a fixed linear leak between neighboring modules before that fit. Nothing is trained offline.

A. Calibration from function patterns

Every photograph supplies its own labelled samples. The QR specification fixes the position of the finder, alignment and timing patterns, and both symbol designs fix their colors, so a decoder knows the true color of these modules before reading any data. The calibration set is the three 8×8 finder blocks, including their white separators, and every 5×5 alignment pattern. For qrism symbols the two timing lines are added, because version 1 has no alignment pattern and its finders contain no black module.

The resulting classes are unbalanced. In a HiQ photograph black appears only in one alignment pattern and contributes 17 samples, against several hundred for white. This is the main reason the classifier below regularizes its covariances.

B. Inter-module leak inversion

Cross-module interference is modeled as a symmetric leak between each module and its edge neighbors. With $\mathcal{N}(p)$ the up to four edge neighbors of p inside the grid, $n_p = |\mathcal{N}(p)|$, and $\mathbf{t}_p$ the value the module would have without leakage,

$$\mathbf{m}_p = (1 - n_p\beta)\,\mathbf{t}_p + \beta \sum_{q \in \mathcal{N}(p)} \mathbf{m}_q, \tag{1}$$

where β is the fraction each neighbor contributes. Writing the neighbors' term with their measured rather than true values makes the model invertible in one pass:

$$\hat{\mathbf{t}}_p = \frac{\mathbf{m}_p - \beta \sum_{q \in \mathcal{N}(p)} \mathbf{m}_q}{1 - n_p\beta}. \tag{2}$$

The correction acts on the module grid rather than on pixels, costs four reads per module, and handles the symbol border by using each module's actual neighbor count. Inside a uniform region it returns the input unchanged, so it alters only modules next to a color change, which is where interference occurs. The gain $1/(1 - 4\beta)$ amplifies sensor noise and diverges at $\beta = 0.25$, so β trades interference removed against noise added. We use one fixed $\beta = 0.06$ for every photograph and every channel; Section VI shows how it was chosen and how it varies with module size and lighting.

Prior treatments of cross-module interference are heavier. HiQ's QDA-CMI and LSVM-CMI learn neighbor weights offline jointly with the classifier [4], and Bagherinia and Manduchi decode blurred color patches by iterative joint estimation [7]. Equation (2) has one parameter, needs no training data and runs in a single pass.

C. Regularized discriminant analysis

After correction, the calibration samples of each color c give a centroid $\boldsymbol{\mu}_c$, taken as the per-channel median, and a covariance Σ_c about it. The pooled within-class covariance Σ is the scatter of all calibration samples about their own class centroids. Each class covariance is shrunk toward the pooled one,

$$\Sigma_c(\lambda) = (1 - \lambda)\,\Sigma_c + \lambda\,\Sigma, \tag{3}$$

following the first of the two regularizations of Friedman [29]. A class with fewer than eight samples uses Σ directly, and a ridge of 10^{-3} times the mean variance keeps every matrix invertible. A module is assigned the color minimizing

$$g_c(\hat{\mathbf{t}}_p) = (\hat{\mathbf{t}}_p - \boldsymbol{\mu}_c)^\top \Sigma_c(\lambda)^{-1} (\hat{\mathbf{t}}_p - \boldsymbol{\mu}_c) + \ln |\Sigma_c(\lambda)|, \tag{4}$$

the Gaussian discriminant with equal priors. With $\lambda = 1$ this is linear discriminant analysis (LDA), and with $\lambda = 0$ quadratic discriminant analysis (QDA). We use $\lambda = 0.5$.

The learned metric is what separates this classifier from nearest-centroid rules. Shading and exposure changes move all three channels together: in five CUHK-CQRC photographs we inspected, the pooled within-class correlations between channels are all positive, mostly between 0.4 and 0.8. A Euclidean distance cannot tell that drift from a change of color, which is why dark blue modules in shadow are read as black. The inverse covariance discounts the shared-brightness direction and weights differences across it.

The calibration samples are taken from the corrected grid (2), so the classifier is fitted in the space it is applied in. Per photograph the fit estimates eight centroids and covariances from a few hundred samples and inverts eight 3×3 matrices; per module it evaluates eight quadratic forms. On one core of an Apple M4 the whole color recovery stage, fit

TABLE I
THE QRISM DATASET BY CAPTURE CONDITION: PHONES USED, MODULE SIZES OF THE SOURCE RENDERS IN PIXELS, PHOTOS, AND PHOTOS THE LOCALIZER FOUND AT THE RIGHT VERSION.

Condition	Phones	px	Photos	Localized
Reference capture	4	12, 20	1120	1024
Outdoor daylight	1	12, 20	280	271
Phone flash on	2	12, 20	560	458
Shown on a display	1	12, 20	280	75
Oblique view	1	12, 20	40	0
Close range	1	20	28	28
8 px modules	2	8	280	259
White balance cool	1	12, 20	280	23
White balance neutral	1	12, 20	280	145
White balance warm	1	12, 20	280	173
Total	4		3428	2456

included, takes 62.3 ns per module, or 1.95 ms for a version-40 symbol of 177×177 modules. Leak inversion accounts for 6.7 ns of this, against 55.6 ns for RDA alone. Image decoding and localization, which every pipeline shares, are not included.

V. EXPERIMENTAL SETUP

A. Datasets

CUHK-CQRC [4] contains photographs of printed three-layer HiQ symbols at versions 21, 25, 30, 35 and 40, taken with several phones under different lighting. Our copy holds the 5544 images of the public release in eight capture folders. The release's metadata marks 5381 of them as HiQ's test set, 65 as the images HiQ's models were trained on, and 98 as unused; the HiQ paper reports 5390 test samples [4]. Our method trains nothing, so its results use every image; the comparison with the HiQ decoder uses the test set alone. One image appears twice under two names. Ground truth for each image is the clean source render of its version.

The *qrism dataset* is new and will be released with this paper. It contains 3428 photographs of qrism symbols at versions 1, 5, 10, 15, 25, 30 and 40, each at all four error-correction levels. Symbols were rendered at 12 and 20 pixels per module, or 8 in the small-module condition, and each was photographed five times under the conditions of Table I with four phones: Apple iPhone 13 and iPhone 16, Motorola Edge 50 Fusion and OnePlus Nord CE [TODO: exact OnePlus model]. [TODO: print setup: printer, paper, printed symbol size, capture distance; display used for the on-screen condition; what distinguishes the close and reference captures]

B. Protocol

Every color recovery pipeline receives identical input: the same localization and the same one sample per module, from the library's localizer (Section III). Only stage (3) of the pipeline differs between rows of a table. A photograph that the localizer misses, or reads at a version with no source render, is excluded from every color recovery rate and counted separately. On CUHK-CQRC 4600 of 5544 photographs are localized (929 not found, 15 read at a wrong version); on the qrism dataset 2456 of 3428 are (972 not found or read at a wrong version). All rates on a dataset therefore share one denominator, which lets two settings be compared photograph by photograph.

A predicted HiQ grid is split into its three layers, each layer is rendered as a monochrome symbol and decoded with the library's standard QR reader, and the symbol counts as decoded when all three messages equal those of the source. For a predicted qrism grid the format information must match the source; the three channels are then decoded with their Reed-Solomon blocks and the message compared with the source.

C. Metrics and statistics

Module accuracy is the share of non-function modules whose color is right; function modules are left out because they overlap the calibration set. The *bit error rate* (BER) is the share of layer bits that are wrong before Reed-Solomon correction, over the same modules, as reported by HiQ [4]. The *layer decode rate* counts HiQ layers whose message is right. The *symbol decode rate* counts symbols decoded in full; HiQ's decoding failure rate is 100 minus this rate.

Decode rates carry 95% Wilson intervals [42]. Two settings on the same photographs are compared with an exact McNemar test [43] on the photographs that one decodes and the other does not. The two parameters, β and λ, are chosen by symbol decode rate on each dataset separately, and we report whether the choice made on one dataset holds on the other.

D. Baselines

The color recovery baselines all use the same calibration modules as our method. *Nearest centroid* assigns the closest calibration median in raw RGB, in optical density relative to the white calibration modules, or after a per-channel stretch that maps the black and white medians to 0 and 1. *Interference cancellation* follows Blasinski et al. [3]: a non-negative 3×3 mixing matrix is fitted to the calibration samples in optical density or absorptance and inverted, and each recovered channel is thresholded midway between the nearest on and off calibration colors, or with a local block threshold. A *local threshold* on the raw channels mirrors monochrome QR binarization. On qrism symbols we also run the hard-decision classifier of JAB Code [5] with its image balancing, using one palette of calibration medians because no qrism finder carries all eight colors. *LDA* and *QDA* are the $\lambda = 1$ and $\lambda = 0$ ends of our classifier without leak inversion.

As an end-to-end reference we run the released HiQ decoder [4] with its shipped models: QDA, QDA with cross-module interference correction (CMI), and a layered SVM. In the released code the CMI step keeps only the center module's term and drops the four neighbor terms, so QDA with and without CMI give identical results; we also report a corrected version that applies all five terms. HiQ uses its own localization and sampling, so this comparison is of complete decoders, not of color recovery alone (Section VI-E).

TABLE II

COLOR RECOVERY PIPELINES, ALL FED THE SAME MODULE SAMPLES: BIT ERROR RATE BEFORE REED-SOLOMON CORRECTION AND SYMBOL DECODE RATE, IN PERCENT, OVER THE PHOTOGRAPHS LOCALIZED AT THE RIGHT VERSION (4600 ON CUHK-CQRC, 2456 ON QRISM). CENTROID: NEAREST CALIBRATION CENTROID; INTERF. CANC.: INTERFERENCE CANCELLATION; OURS: RDA WITH LEAK INVERSION. OURS DECODES MORE SYMBOLS THAN EVERY BASELINE, WITH EXACT MCNEMAR $p < 10^{-21}$.

	CUHK-CQRC		qrism	
Pipeline	BER	Symbol	BER	Symbol
Centroid, RGB	5.90	19.9	7.18	16.4
Centroid, optical density	7.31	12.4	10.30	5.8
Centroid, black/white stretch	5.96	20.0	7.19	16.5
Interf. canc., density	6.91	14.3	7.92	18.5
Interf. canc., absorptance	6.25	14.6	–	–
Interf. canc., local thr.	6.75	18.0	–	–
Local threshold	16.61	0.0	–	–
JAB Code classifier	–	–	8.89	13.2
LDA	5.54	32.2	6.23	22.3
QDA	6.10	21.2	9.21	12.9
RDA ($\lambda = 0.5$)	5.29	35.4	5.90	24.3
Ours, β fit per image	5.00	39.3	–	–
Ours, $\beta = 0.06$	**4.61**	**42.9**	**4.85**	**33.9**

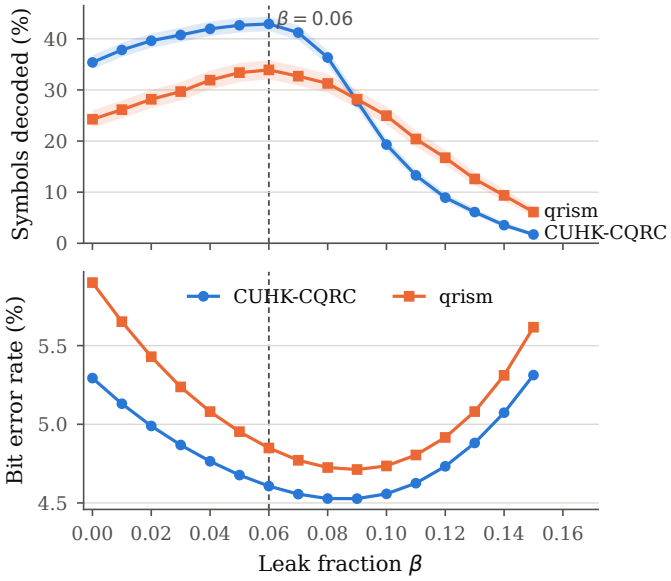


Fig. 1. Leak fraction β against symbol decode rate with its 95% Wilson band (top) and pre-correction bit error rate (bottom), RDA shrink $\lambda = 0.5$. $\beta = 0$ is RDA without leak inversion. Each dataset's photographs are those localized at the right version.

VI. RESULTS

A. Color recovery pipelines

Table II compares color recovery pipelines on both datasets. Nearest centroid in raw RGB decodes 19.9% of symbols on CUHK-CQRC and 16.4% on qrism, and per-colorant interference cancellation [3] 14.3% and 18.5%. JAB Code's classifier decodes 13.2% on qrism. Keeping the same centroids but measuring distance with the RDA metric raises the rate to 35.4% and 24.3%, and adding leak inversion at $\beta = 0.06$ raises it to 42.9% (95% CI 41.5–44.3) and 33.9% (32.1–35.8). Every baseline decodes fewer symbols than our method on both datasets; the closest has exact McNemar $p < 10^{-21}$. Leak inversion alone decodes 399 CUHK-CQRC photographs that RDA misses and loses 53 ($p < 10^{-65}$); on qrism the counts are 310 and 73 ($p < 10^{-35}$).

Module accuracy understates these differences. Nearest centroid and our method differ by three points in module accuracy (87.0% against 90.1%) but by more than a factor of two in symbols decoded, because a symbol fails as soon as one Reed-Solomon block in one layer holds more errors than it can correct. We therefore rank pipelines and choose parameters by symbol decode rate.

Choosing β per photograph does worse than the fixed value: minimizing the spread of the corrected grid around its palette gives 39.3%, and an optical estimate of blur measured on the finder edges gives 36.3%.

B. Leak fraction

Fig. 1 sweeps β from 0 to 0.15. On each dataset separately the decode rate peaks at $\beta = 0.06$: 42.9% on CUHK-CQRC against 35.4% without leak inversion, and 33.9% on the qrism dataset against 24.3%. Because both datasets select the same value, choosing β on one and testing on the other costs nothing. The optimum is narrow but not sharp: $\beta = 0.05$ cannot be told apart from 0.06 ($p = 0.294$ and $p = 0.246$), while 0.04 ($p = 0.002$, $p < 10^{-3}$) and 0.07 ($p < 10^{-10}$, $p = 0.005$) are worse on both datasets. Past 0.08 the noise gain dominates and decoding collapses, to 8.9% and 16.7% at $\beta = 0.12$.

The bit error rate does not peak where decoding does. At $\beta = 0.06$ it falls from 5.29% to 4.61% on CUHK-CQRC and from 5.90% to 4.85% on qrism, but it keeps falling, to $\beta = 0.09$ on CUHK-CQRC and 0.09 on qrism. Our reading is that the photographs that can be decoded at all want less correction than those that cannot. On the qrism dataset the bit error rate of 20 px symbols is lowest at $\beta = 0.05$, next to their decode optimum, while 12 px and 8 px symbols, which rarely decode, are best served by $\beta = 0.11$ and 0.15. Those symbols pull the pooled bit error rate toward larger β without adding decoded symbols. The per-photograph optical estimate shows the same split: it has the lowest bit error rate of any setting on both datasets, yet on CUHK-CQRC it decodes fewer symbols than the fixed value (36.3% against 42.9%, $p < 10^{-50}$); on qrism the two tie (34.0% against 33.9%, $p = 1.000$).

Blue is the weakest channel on both datasets. At $\beta = 0.06$ its bit error rate is 6.63% on CUHK-CQRC against 3.63% and 3.56% for red and green, and 7.39% on qrism against 2.21% and 4.95%. HiQ reports the same for its third layer [4]. Since a qrism symbol decodes only when all three channels do, the blue channel bounds its decode rate.

TABLE III
SYMBOL DECODE RATE (%) AGAINST THE RDA SHRINK λ, WITHOUT LEAK INVERSION ($\beta = 0$) AND WITH IT ($\beta = 0.06$).

	CUHK-CQRC		qrism	
λ	$\beta = 0$	0.06	$\beta = 0$	0.06
0.0 (QDA)	21.2	28.6	12.9	17.8
0.2	33.6	39.5	21.5	31.6
0.4	35.5	42.5	23.5	33.3
0.5 (used)	**35.4**	**42.9**	**24.3**	**33.9**
0.6	34.9	43.1	24.6	33.9
0.8	33.7	42.8	24.3	33.0
1.0 (LDA)	32.2	41.9	22.3	30.0

TABLE IV
QRISM DATASET: SYMBOL DECODE RATE (%) BY MODULE SIZE AND CAPTURE CONDITION, WITHOUT LEAK INVERSION, AT $\beta = 0.06$, AND AT THE BEST FIXED β FOR THAT GROUP. n IS LOCALIZED PHOTOS.

Group	n	$\beta = 0$	0.06	best β	at best
8 px	259	0.0	0.4	0.10	3.9
12 px	1061	9.3	18.5	0.08	19.8
20 px	1136	43.8	56.0	0.06	56.0
regular	1024	27.8	39.8	0.05	40.5
daylight	271	46.1	41.3	0.01	49.1
flashlight	458	14.0	28.4	0.10	31.0
monitor	75	90.7	89.3	0.01	92.0
close	28	82.1	85.7	0.02	85.7
wb-cool	23	30.4	47.8	0.10	65.2
wb-neutral	145	11.7	35.9	0.10	55.2
wb-warm	173	4.0	16.2	0.06	16.2

C. Covariance shrink

Table III crosses the RDA shrink λ with β. Decode rates are flat for λ between 0.4 and 0.7 and fall at both ends. With leak inversion, QDA decodes 28.6% on CUHK-CQRC ($p < 10^{-173}$ against $\lambda = 0.5$) and 17.8% on qrism ($p < 10^{-107}$): per-class covariances estimated from few samples (17 for black on CUHK-CQRC), then amplified by the correction, generalize poorly. LDA loses less, 41.9% ($p < 10^{-3}$) and 30.0% ($p < 10^{-13}$). The best cell of the full grid is $\beta = 0.06$, $\lambda = 0.6$ on CUHK-CQRC (43.1%, $p = 0.211$ against $\lambda = 0.5$) and $\beta = 0.06$, $\lambda = 0.5$ on qrism. For every $\lambda \geq 0.3$ the best β is 0.05 to 0.06, so the two parameters can be set independently.

D. Module size and capture condition

Table IV breaks the qrism results down. The best β grows as modules shrink: 0.06 at 20 px, 0.08 at 12 px and 0.10 at 8 px. This is what a blur of fixed width in pixels predicts, since it covers a larger fraction of a smaller module. At 8 px almost nothing decodes, at most 3.9%. Capture conditions move the optimum as well: daylight photographs do best at $\beta = 0.01$ (49.1% against 41.3% at 0.06), while the neutral white-balance condition does best at 0.10 and the flash condition at 0.10. These per-group optima are chosen on the photographs they are scored on, so they overstate what a per-group β would gain; they show that a single β is a compromise across conditions.

TABLE V
END-TO-END DECODERS ON HIQ'S TEST SPLIT OF CUHK-CQRC, PAIRED BY PHOTO. EACH DECODER USES ITS OWN LOCALIZATION AND SAMPLING. DISC.: PHOTOS ONLY OURS DECODED / ONLY HIQ DECODED; EVERY DIFFERENCE HAS EXACT MCNEMAR $p < 10^{-127}$.

Decoder	Symbol %	95% CI	Disc.
Localized by both ($n = 3404$)			
Ours, $\beta = 0.06$	50.1	[48.4, 51.7]	
HiQ QDA	26.0	[24.6, 27.5]	926/108
HiQ QDA + CMI, corrected	29.5	[28.0, 31.0]	824/123
HiQ layered SVM	25.5	[24.1, 27.0]	931/95
All photos ($n = 5381$)			
Ours, $\beta = 0.06$	35.3	[34.0, 36.5]	
HiQ QDA	17.2	[16.2, 18.3]	1119/149
HiQ QDA + CMI, corrected	19.5	[18.4, 20.5]	1017/167
HiQ layered SVM	16.7	[15.7, 17.7]	1124/126

E. Comparison with the HiQ decoder

This comparison uses only HiQ's test split, so no photograph HiQ's models were trained on is scored. Of its 5381 photographs the HiQ decoder localizes 3812 and ours 4451; 3441 are localized by both, 3404 of them at the same version. Table V compares the decoders on these photographs and on all photographs, counting a photograph that a decoder does not localize as a failure. On the common set our decoder reads 50.1% of symbols against 29.5% for HiQ's best configuration, QDA with the corrected CMI. It leads in all 5 symbol versions and all 8 capture folders. Without leak inversion it reads 41.7%, still ahead of HiQ.

The HiQ classifiers were also timed on the same machine, over the same span as ours (Section IV): 53.7 ns per module for the layered SVM, 136.8 ns for QDA and 161.1 ns for QDA with the corrected CMI, against 62.3 ns for ours. These figures do not isolate the algorithms. HiQ runs as JIT-compiled Java and ours as Rust, which favors ours; HiQ fits nothing per photograph while our time includes the fit, which favors HiQ. They do show that the costs are of the same order: ours falls between HiQ's layered SVM and its QDA without any training, and the layered SVM, the only faster one, decodes 25.5% of the common set.

One qualification applies. Each decoder uses its own localization and module sampling, so the gap combines color recovery and sampling; a comparison of the classifiers alone needs HiQ's classifier run on our samples. [TODO: run HiQ's QDA and corrected CMI on the library's samples, or state the claim as end-to-end only]

VII. LIMITATIONS

A single β is a compromise. The best value depends on module size in pixels and on lighting (Table IV), and the per-photograph estimates we tried do not reliably beat the fixed value. Estimating β from a photograph in a way that tracks decodability rather than bit error rate is open.

Calibration assumes the function patterns are intact. Glare, damage or occlusion on a finder or alignment pattern corrupts the samples the whole photograph is fitted on.

Symbols with 8 px modules almost never decode, and none of the oblique-view photographs is localized, so neither condition informs the comparison of color recovery methods.

Each module is read from one pixel at its projected center. Averaging several pixels may reduce noise but would also mix in more of the neighbors that leak inversion removes; we have not measured this trade-off.

The comparison with HiQ is end to end (Section VI-E) and does not separate color recovery from localization and sampling.

VIII. Conclusion

[TODO: write last: restate the method in two sentences, the headline numbers (42.9% and 33.9% symbol decode rate at $\beta = 0.06$, $\lambda = 0.5$, chosen on each dataset and agreeing), the release of the dataset and code, and the open problem of choosing β per photograph.]

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